

Question Paper

Exam Date & Time: 21-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATION - DECEMBER 2025

Industrial Nanotechnology [PHY_4344]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Identify how the electronic properties of nanomaterials change with size reduction and how this affects nanodevice performance. (3)
A) Course Outcome: "CO1" / Cognitive Level: "Application"
B) Assess how nanomaterials improve the efficiency of solar cells and batteries. (3)
Course Outcome: "CO1" / Cognitive Level: "Application"
C) Discover the application of BioMEMS in neural prosthetics and brain-machine interfaces. (4)
Course Outcome: "CO1" / Cognitive Level: "Analysis"
- 2) Find out the ideal properties of a photosensitizer used in photodynamic therapy. (3)
A) Course Outcome: "CO2" / Cognitive Level: "Application"
B) Assess how quantum dots and gold nanoparticles can be used for targeted drug delivery. (3)
Course Outcome: "CO2" / Cognitive Level: "Application"
c) Analyze how stimuli-responsive materials can be used to control drug release based on pH or temperature. (4)
Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 3) Identify the application of quantum dots in bio-imaging and medical diagnostics. (3)
A) Course Outcome: "CO3" / Cognitive Level: "Application"
B) Find out how zeolites can be utilized as slow-release fertilizers and identify the advantages they offer compared to conventional fertilizers. (3)
Course Outcome: "CO3" / Cognitive Level: "Application"
C) Analyze how the particle size of a nanocatalyst affects its catalytic activity, selectivity, and stability. (4)
Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) Identify how predictive analytics helps farmers make better decisions regarding planting and (3)

A) harvesting schedules.

Course Outcome: "CO4" / Cognitive Level: "Application"

B) Find out how smart delivery systems can help maintain soil fertility and reduce nutrient leaching. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

C) Categorize traditional processing methods with modern, technology-driven methods in terms of product quality and cost-effectiveness. (4)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

5) Assess how electrospinning is used to produce polymer nanofibers and discuss its applications. (3)

A) Course Outcome: "CO5" / Cognitive Level: "Application"

B) Find out how combining nanoparticles with hydrogels or nanofibers can enhance the mechanical strength and bioactivity of scaffolds. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

C) Analyze how nano-coatings are used to impart water-repellent and stain-resistant properties to fabrics. (4)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

PHY_4344:Industrial Nanotechnology

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

PHY_4344:Industrial Nanotechnology

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	10	30(60%)
4	Analysis	5	20(40%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)